Lecture plan · for the lecturer

Teaching statistical learning at university: unit check

Data science · University · Unit check

Time	Materials
40 min	Projector or board, and the lecture slides.

Learning goals

- Sampling and the bootstrap
- Bias, variance and CV
- Causal inference
- Module design

Activity: Peer-instruction review

1. Vote (10 min): Show each question; students answer alone and commit to an answer.
2. Discuss (15 min): Where answers split, pairs argue for two minutes and vote again.
3. Explain (10 min): Work through the questions most students missed, showing the reasoning step by step.
4. Plan (5 min): Note which misconception to revisit next week and how.

Discussion

- Which misconception will you target first?
- Which lab will you run this term?
- Which assessment will you change?

Exit ticket

- One change I will make to my module this term, and how I will know it worked.

Try it in your next lecture
Bring your redesigned alignment table to the next department meeting and ask one colleague to review it.

Seminar sheet · for students

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Name _____ Class _____ Date _____

1 **σ = 20 and n = 100. What is the standard error of the mean?**

◯ 2 ◯ 0.2 ◯ 20 ◯ 10

2 **A 95% CI of 60–68 means P(60 < μ < 68) = 0.95.**

◯ True ◯ False

3 **Which are limits of the bootstrap? Choose all that apply.**

☐ It cannot fix a biased sample ☐ It is unreliable for the maximum

☐ It only works for the mean

4 **Work out the CV error.**

Fold errors 6, 8, 7, 9, 10 give a CV error of _____ .

Word bank: 8 · 40 · 10

5 **Put a leak-free workflow in order.**

☐ Report error once on the test set

☐ Choose the model by CV error

☐ Inside each CV fold, fit preprocessing and model

☐ Hold out a final test set